

Discrete fixed-target stability regions in the (k_p, dt) plane

Regions follow from $e' = (1 - k_p dt) e$: $|1 - k_p dt| < 1$ iff $0 < k_p dt < 2$ (Reduce, exact). The code

enforces the stricter monotone region $k_p dt \leq 1$. Valid only for a fixed target, unsaturated control, and no safety projection.

Classification: Symbolically verified under stated assumptions (fixed target, no saturation/projection)

main 93745582d849 | Wolfram 15.0.0 | WorkingPrecision 50 | cross-language tol 1e-9 | exact region boundaries

